

Hyperwelcome

Theodor Holm Nelson

To cite this article: Theodor Holm Nelson (1989) Hyperwelcome, Hypermedia, 1:1, 3-5, DOI: [10.1080/09558543.1989.12031154](https://doi.org/10.1080/09558543.1989.12031154)

To link to this article: <https://doi.org/10.1080/09558543.1989.12031154>



Published online: 29 Oct 2018.



Submit your article to this journal [↗](#)



View related articles [↗](#)



Citing articles: 3 View citing articles [↗](#)

HYPERWELCOME

THEODOR HOLM NELSON

Autodesk, Inc., 2320 Marinship Way, Sausalito CA 94965, USA

It is of course inevitable. It should have been obvious to anyone that general interactive media would appear and proliferate. Text, graphics, audio and video can now come alive in unified, responding, explorable new works that present facts and ideas: hypermedia. Unlimited new forms of connection and branching now offer the chance to explore ideas — to follow different lines of thought, different forms of exposition, different connections in a subject, in ways never before possible. The sequential writings and media of the past have given us only the dimmest precedents.

Yet all roads have led to this. This new hypermedia synthesis does not come merely out of computer technology; it comes from literature and the printing arts, evolving toward new forms; from movies, at last giving the viewer more detailed choices than whether to come or go; from instructional technology, (especially as it sheds its linear and curricular biases); from board games, from video games and adventure games; and from the imaginations of thousands of people everywhere.

So many thoughts of our time have begun, 'If we can put a man on the moon . . .' Similarly, if we can put a picture on a responding screen, why cannot that screen do practically anything?

And so it can. And so these new techniques are springing up everywhere, in minds and on screens around the world.

Of course, the many people who come up independently with the hypermedia idea tend also to have other ideas already, and so are divided into sects like people in any other field. Each sect brings its own presumptions; now we hear a thousand new vying terms and rules, based in the different disciplines and points of view from which people come. We hear of 'opening screens' versus 'home cards,' menus pull-down versus tear-off, macros versus scripts, webs versus docuverses, anchors versus endsets, courseware versus show biz.

There are also special doctrinal positions to note. Enthusiasts of CD-ROM, for example, are preoccupied with specific equipment and the possible effects of its wide adoption. But the problem with being equipment-driven is that the equipment gets to decide where you are going. This is rather like letting the car decide where to go by taking your hands off the steering wheel, on the presumption that the car has more right to decide where to go than you do (however temporary and shallow may be the car's concerns and momentum). (No one now among us has done this, I think, and I do not recommend it.)

Then there is the matter of Artificial Intelligence, that anomalous new discipline and/or religious myth which so many technological sympathisers espouse. Like the CD-ROM enthusiasm it suggests that machines ought to be in command rather than we.

At AI's most religious level, believers think it is our duty to create a master race of robots or programs that will replace us. (Fortunately this guiding ideal does not necessarily affect practical exertions or day-to-day plans.) But still, many expect Artificial Intelligence to relieve us of various kinds of problems and decisions, whether we want to be relieved of them or not. In today's scrambled media scene, so full of ill-matching and confused setups, many say we will use artificial intelligence as a ketchup to disguise bad interactive designs, just as ketchup is sometimes used to disguise unappetizing food.

But we need judge none of these matters now. The natural uses of these things will become evident. We should not disdain any enthusiasms, however benighted, for it is by enthusiasm that we survive and achieve.

However, not everyone is enthusiastic about the coming hypermedia world. Till now, linear writing and tabular lists have been a way to represent and tame the world of interconnection. Linear writing and tabular lists will in some ways now give way to a new world, nonlinear and non-tabular, and some people will find this distressing. (Marshall McLuhan himself told me that he preferred a linear world, and saw the nature of civilization as profoundly linear and representable as tables and lists).

Some people are distressed at the new variety of interconnectedness this makes it possible to represent. But the interconnectedness is there, has been there all along; it is just that till now we had tamed it, made it seem flat by making our lists and straightening it into our descriptive sequences.

But now we can place doorways on these sequences. Now we can represent the anatomy of an interconnection, not just by a description, but in a way that people can follow, opening the door into the thing itself. Perhaps it is this that disconcerts. It is one thing to say that there exists a labyrinth, and another to make the labyrinth present with all its variety of choice.

Another thing that bothers many people about the interactive object is where does it end? How do you know when you're done? With a book or an article you achieve perfunctory completion by reading from the beginning to the end. Hypermedia make completion more a matter of conscience than of a certifiable terminus. This may be disconcerting to those less accustomed to the uses of conscience.

The world of academia will have its own problems with

hypermedia. One problem, of direct importance to our contributors, is that an interactive object or production is not like a conventional publication. Yes, the author can put it on her or his curriculum vitae, but cannot make a photocopy of it to send along. The work must reside in some other form than paper, and come to life through mechanisms and controls that are not yet standard. And thus when you have created something in such a medium, you can only tell people how to view and use it, but you cannot yet give them a copy that is useful in itself.

And this is of course the problem of order and compatibility. Now there are on the market dozens of hypertext and hypermedia systems, showing little inclination toward compability of any kind. Thus the question becomes not merely what system to use, but what system to use that will last? The problem of Balkanization, of incompatibility between mutually exclusive worlds, is distressing to anyone who seeks to create or participate in any permanence or generality.

Further, how is it possible to tie together works in hypermedia? What overall environment will make it possible to connect separate hypermedia works together, let alone create a pool of materials comparable to the pooling of books in libraries as we know them?

These are serious issues. My own proposal for a unifying docuverse of xanalogical storage,¹ combining mutual linkage and quotation in a unified hypermedia field, is the only proposal I know of at present to unite the so-far incompatibly different hyperworlds.

Another way to unify the hypermedia is to talk about them in a common forum. And so this journal: limited, unfortunately, to the common denominator of paper. Imagine! Here are we at the threshold of a new age of interactive text, graphics and audio and of course, here am I still writing about it in a linear fashion! Let us enjoy the moment and strike while the irony's hot. And enjoy as well this journal; for it is time to celebrate what is to come, little though we know what that may be.

REFERENCES

1. NELSON, T.H. *Literary machines*. Published by the author, 1981; various subsequent editions.